

1. (a) For each of the following multiple-choice questions, select the correct answer.

Note: in the following multiple-choice questions that +1 mark will be awarded for each correct answer, -0.5 for each incorrect answer and 0 marks for any answer left blank:

(i) The Computer Misuse Act...

- a. Never considers whether the offender intended to commit a crime.
- b. Always considers whether the offender intended to commit a crime.
- c. May consider whether the offender intended to commit a crime.

(ii) A scientific approach...

- a. Compares observations with hypotheses.
- b. Compares observations with predictions.
- c. Compares hypotheses with predictions.

(iii) When considering the generalisability of a human computer interaction experiment...

- a. Generalisability is increased by changing the definition of the independent variable.
- b. We give up some control in order to increase generalisability
- c. Changing the definition of the random variables can increase generalisability.

(iv) An annotated bibliography...

- a. Must include all details of the paper.
- b. Only contains peer reviewed research articles.
- c. Is often included at the end of an article.

(v) When using information from a research paper in a document intended for the purposes of assessment...

- a. Plagiarism can refer to using the same ideas in the paper and presenting them as your own.
- b. You can use the exact words used in the paper provided that you quote them in quotation marks.
- c. You do not need to give a citation if you use a formula from a research paper.

[5]

(b) Discuss four reasons why societal issues in IT often require existing legislation be amended or new legislation be introduced.

[4]

(c) A video game development company have just finished implementing a character creator system to allow players to customise the avatar the player uses within the game.

One designer suggests the company attempt to copyright this idea so it can only be used in games created by the company.

Describe one reason why the company's attempt to copyright this might be rejected.

[2]

(d) A researcher is conducting an experiment to understand how people use a self-checkout in a supermarket.

The researcher has chosen to observe users interacting with the system in real-time in the supermarket and to interview some users about their experience using self-checkout machines.

Describe 5 reasons why the researcher might have chosen to use both methods as part of their investigation.

[5]

(e) A researcher has developed a new speech recognition algorithm for a smart home digital assistant.

The researcher believes the new speech recognition algorithm ("SuperSpeech") provides better speech recognition performance than the existing algorithm ("SimpleSpeech").

The researcher is designing experiment (with human participants) to investigate this belief.

Identify one quantitative and one qualitative dependent variable for this experiment. Justify your choice of each chosen variable.

[4]

- 2.** A researcher is evaluating a new smartphone app designed to help guide people through a bodyweight workout using video instructions.

The researcher has planned a simple experiment to test if the app makes it easier to guide people through a bodyweight workout compared to paper-based instructions.

First, the researcher will ask participants to go through a fixed workout using the paper-based instructions.

Next, the researcher will ask participants to go through the workout again using the new smartphone app (the video instructions approach).

The researcher will measure the time taken to complete the workout each time, to see if the average completion time is lower when using the smartphone app. If it is lower, the researcher will conclude that the smartphone app makes it easier to guide users through a workout than paper-based instructions.

- (a)** Critique this experiment by explaining three positive aspects of its design.

[6]

- (b)** Critique this experiment by explaining four negative aspects of its design. Propose changes to address any identified negative issues.

[12]

- (c)** Describe two reasons why inductive inference would be used to evaluate the hypothesis of this experiment opposed to deductive inference.

[2]

- 3.** You have developed a new user interface for a spreadsheet editor.

You believe your new user interface (“Training Interface”) will increase the performance of novice users of the spreadsheet editor compared to the current system (“Original Interface”).

You plan to conduct a research project with an experiment (with human participants) to investigate this belief.

- (a)** Propose a hypothesis for an experiment to investigate this belief.

[2]

- (b)** Propose an experiment to investigate this hypothesis.

You should cover all aspects of a research project, except performing a literature survey and writing a research proposal.

[18]